UIL COMPUTER SCIENCE WRITTEN TEST

2026 STATE

MAY 2026

General Directions (Please read carefully!)

1. DO NOT OPEN THE EXAM UNTIL TOLD TO DO SO.
2. There are 40 questions on this contest exam. You will have 45 minutes to complete this contest.
3. All answers must be legibly written on the answer sheet provided. Indicate your answers in the appropriate blanks provided on the answer sheet. Clean erasures are necessary for accurate grading.
4. You may write on the test packet or any additional scratch paper provided by the contest director, but NOT on the answer sheet, which is reserved for answers only.
5. All questions have ONE and only ONE correct answer. There is a 2-point penalty for all incorrect answers.
6. Tests may not be turned in until 45 minutes have elapsed. If you finish the test before the end of the allotted time, remain at your seat and retain your test until told to do otherwise. You may use this time to check your answers.
7. If you are in the process of actually writing an answer when the signal to stop is given, you may finish writing that answer.
8. All provided code segments are intended to be syntactically correct, unless otherwise stated. You may also assume that any undefined variables are defined as used.
9. A reference to many commonly used Java classes is provided with the test, and you may use this reference sheet during the contest. AFTER THE CONTEST BEGINS, you may detach the reference sheet from the test booklet if you wish.
10. Assume that any necessary import statements for standard Java SE packages and classes (e.g., `java.util`, `System`, etc.) are included in any programs or code segments that refer to methods from these classes and packages.
11. NO CALCULATORS of any kind may be used during this contest.

Scoring

1. Correct answers will receive **6 points**.
2. Incorrect answers will lose **2 points**.
3. Unanswered questions will neither receive nor lose any points.
4. In the event of a tie, the student with the highest percentage of attempted questions correct shall win the tie.

STANDARD CLASSES AND INTERFACES – SUPPLEMENTAL REFERENCE

package java.lang

```
class Object
  boolean equals(Object anotherObject)
  String toString()
  int hashCode()

interface Comparable<T>
  int compareTo(T anotherObject)
    Returns a value < 0 if this is less than anotherObject.
    Returns a value = 0 if this is equal to anotherObject.
    Returns a value > 0 if this is greater than anotherObject.

class Integer implements Comparable<Integer>
  Integer(int value)
  int intValue()
  boolean equals(Object anotherObject)
  String toString()
  String toString(int i, int radix)
  int compareTo(Integer anotherInteger)
  static int parseInt(String s)

class Double implements Comparable<Double>
  Double(double value)
  double doubleValue()
  boolean equals(Object anotherObject)
  String toString()
  int compareTo(Double anotherDouble)
  static double parseDouble(String s)

class String implements Comparable<String>
  int compareTo(String anotherString)
  boolean equals(Object anotherObject)
  int length()
  String substring(int begin)
    Returns substring(begin, length()).
  String substring(int begin, int end)
    Returns the substring from index begin through index (end - 1).
  int indexOf(String str)
    Returns the index within this string of the first occurrence of str. Returns
    -1 if str is not found.
  int indexOf(String str, int fromIndex)
    Returns the index within this string of the first occurrence of str, starting
    the search at fromIndex. Returns -1 if str is not found.
  int indexOf(int ch)
  int indexOf(int ch, int fromIndex)
  char charAt(int index)
  String toLowerCase()
  String toUpperCase()
  String[] split(String regex)
  boolean matches(String regex)
  String replaceAll(String regex, String str)

class Character
  static boolean isDigit(char ch)
  static boolean isLetter(char ch)
  static boolean isLetterOrDigit(char ch)
  static boolean isLowerCase(char ch)
  static boolean isUpperCase(char ch)
  static char toUpperCase(char ch)
  static char toLowerCase(char ch)

class Math
  static int abs(int a)
  static double abs(double a)
  static double pow(double base, double exponent)
  static double sqrt(double a)
  static double ceil(double a)
  static double floor(double a)
  static double min(double a, double b)
  static double max(double a, double b)
  static int min(int a, int b)
  static int max(int a, int b)
  static long round(double a)
  static double random()
    Returns a double greater than or equal to 0.0 and less than 1.0.
```

package java.util

```
interface List<E>
class ArrayList<E> implements List<E>
  boolean add(E item)
  int size()
  Iterator<E> iterator()
  ListIterator<E> listIterator()
  E get(int index)
  E set(int index, E item)
  void add(int index, E item)
  E remove(int index)

class LinkedList<E> implements List<E>, Queue<E>
  void addFirst(E item)
  void addLast(E item)
  E getFirst()
  E getLast()
  E removeFirst()
  E removeLast()

class Stack<E>
  boolean isEmpty()
  E peek()
  E pop()
  E push(E item)

interface Queue<E>
class PriorityQueue<E>
  boolean add(E item)
  boolean isEmpty()
  E peek()
  E remove()

interface Set<E>
class HashSet<E> implements Set<E>
class TreeSet<E> implements Set<E>
  boolean add(E item)
  boolean contains(Object item)
  boolean remove(Object item)
  int size()
  Iterator<E> iterator()
  boolean addAll(Collection<? extends E> c)
  boolean removeAll(Collection<?> c)
  boolean retainAll(Collection<?> c)

interface Map<K,V>
class HashMap<K,V> implements Map<K,V>
class TreeMap<K,V> implements Map<K,V>
  Object put(K key, V value)
  V get(Object key)
  boolean containsKey(Object key)
  int size()
  Set<K> keySet()
  Set<Map.Entry<K, V>> entrySet()

interface Iterator<E>
  boolean hasNext()
  E next()
  void remove()

interface ListIterator<E> extends Iterator<E>
  void add(E item)
  void set(E item)

class Scanner
  Scanner(InputStream source)
  Scanner(String str)
  boolean hasNext()
  boolean hasNextInt()
  boolean hasNextDouble()
  String next()
  int nextInt()
  double nextDouble()
  String nextLine()
  Scanner useDelimiter(String regex)
```

STANDARD CLASSES AND INTERFACES – SUPPLEMENTAL REFERENCE

Package java.util.function

```
Interface BiConsumer<T,U>
  void accept(T t, U u)

Interface BiFunction<T,U,R>
  R apply(T t, U u)

Interface BiPredicate<T,U>
  boolean test(T t, U u)

Interface Consumer<T>
  void accept(T t)

Interface Function<T,R>
  R apply(T t)

Interface Predicate<T>
  boolean test(T t)

Interface Supplier<T>
  T get()
```

UIL COMPUTER SCIENCE WRITTEN TEST – 2026 STATE

Note: Correct responses are based on **Java SE Development Kit 25 (JDK 25)** from Oracle, Inc. All provided code segments are intended to be syntactically correct, unless otherwise stated (e.g., "error" is an answer choice) and any necessary Java SE 25 Standard Packages have been imported. Ignore any typographical errors and assume any undefined variables are defined as used. **For all output statements, assume that the System class has been statically imported using: `import static java.lang.System.*;`**

Question 1

How many 1s are in the binary representation of the base 10 number 28,520,985?

A) 12 **B)** 13 **C)** 11 **D)** 10 **E)** 14

Question 2

What is output by the code to the right?

A) 400 **B)** 395 **C)** 416 **D)** 415 **E)** 410

```
int x = 123;
int y = 456;
out.println(x + (y % x) - x ^ y);
```

Question 3

What is output by the code to the right?
The next line is a ruler that can be used for your convenience.

```
  01234567890123456789012345678901234567890
```

A) `[      42|-003.14|Hello]`

B) `[42      |-003.14|Hello]`

C) `[      42|  -3.14|Hello]`

D) `[42      |  -3.14|Hello]`

E) `[42      |-003.14|HelloWorld]`

```
int x = 42;
double y = -3.14;
String s = "HelloWorld";

out.printf("[%-8d|%07.2f|%.5s]%n",
        x, y, s);
```

Question 4

What is output by the code to the right?

A) `truetrueHello World`

B) `falsefalseHello World`

C) `truefalseHello World`

D) `truefalse[H,e,l,l,o, ,W,o,r,l,d]`

E) `truetrue[H,e,l,l,o, ,W,o,r,l,d]`

```
String s = "Hello World";
char [] helloWorld = s.toCharArray();
String s2 =
    String.copyValueOf(helloWorld);

out.print(s.equals(s2));
out.print(s == s2);
out.print(helloWorld);
```

Question 5

What is output by the code to the right?

A) `false`

B) `true`

```
boolean a = true;
boolean b = false;
boolean c = a ^ b || a && b;
out.println(c && (a ^ c || b));
```

Question 6

What is output by the code to the right?

A) `2147483647`

B) `-2147483648`

C) There is no output due to a runtime error.

D) There is no output due to a compile error.

```
int x = 0;
int count = 0;

while(x <= Integer.MAX_VALUE) {
    x = Math.incrementExact(x);
    count++;
}
out.println(count);
```

Question 7

What is output by the code to the right?

A) 48 **B)** 191 **C)** 159 **D)** 253 **E)** 155

```
int a = 0b1010_1100;
int b = 0xF3;
int c = 073;
out.println(a & b ^ c | a >> 2 + 1);
```

Question 8

What is output by the code to the right?

A) `Zoo`

B) `ZooWee`

C) `ZooWeeMama`

D) `Wee`

E) `WeeMama`

```
double d = 2.0;
float f = 2.0f;
double d2 = Double.NaN;
float f2 = Float.NaN;
if (d == f)
    out.print("Zoo");
if (d2 == f2)
    out.print("Wee");
if (d2 > f2)
    out.print("Mama");
```

Question 9

What is output by the code to the right?

A) `XYQ61`

B) `XZQ61`

C) `WXYQ63`

D) `XZ71`

E) `XZQ71`

```
int a = 7, b = 2, c = 0;
String r = "";

switch (a % 3 + b) {
    case 4: r += "W"; c = 2;
    case 3: r += "X";
        if (c++ > 0 && --a > 5)
            r += "Y";
        else
            r += "Z";
    case 2: r += "Q"; break;
    default: r += "R";
}
out.println(r + a + c);
```

Question 10

What is output by the code to the right?

A) `0034`

B) `7004`

C) `4034`

D) `7034`

E) `4004`

```
int[] a = {1, 4, 2, 8, 3};
int[] b = a;
int[] c = new int[a.length];

for (int i = 0; i < a.length - 1; i++)
    c[i] = a[i + 1] - a[i];

b = c;
a[0] += b[0] + c[0];
b[0] = 0;

out.println(a[0] + "" + b[0] +
        c[0] + a[1]);
```

Question 11

What is output by the code to the right?

A) `12|5678`

B) `12|168`

C) `12|34134`

D) `12|345678`

E) `46|345678`

```
Scanner sc = new Scanner("12 34\n56 78");
int a = sc.nextInt();
String s = sc.nextLine();
Scanner sc2 = new Scanner(s +
            sc.nextInt() + sc.nextInt());

int x = 0;
while (sc2.hasNextInt())
    x += sc2.nextInt();
out.println(a + "|" + x);
```

Question 12

What is output by the code to the right?

A) `12|9|9`

B) `9|12|9`

C) `5|6|0`

D) `12|6|9`

E) `6|12|5`

```
int s = 0, p = 1, t = 0;
for (int i = 2; i <= 6; i++) {
    switch (i % 4) {
        case 0: s *= i;
        case 1: s += i; break;
        case 2: p *= i;
        case 3: t = s; s = p; p = t;
                break;
    }
}
out.println(s + "|" + p + "|" + t);
```

Question 13

What is output by the code to the right?

A) `3|8` **B)** `3|4`

C) `4|8` **D)** `2|4`

E) `2|8`

```
int a = 5, b = 3, c = 2;
int d = ++a - b--;
out.print(d + "|");
out.print(a > b ? a + c : b * c);
```

Question 14

What is output by the code to the right?

A) `true` **B)** `false`

```
out.println(Double.MIN_VALUE > 0);
```

Question 15

What is the largest possible sum of values in the list?

A) `4 [127, 128, 127] [127, 128]`

B) `0 [127, 128, 127] [127, 128]`

C) `2 [127, 127] [127, 128]`

D) `2 [127, 128, 127] [127, 128]`

E) There is no output due to a runtime error.

```
ArrayList<Integer> a = new ArrayList<>(
    List.of(127, 128, 127, 128));
ArrayList<Integer> b = new ArrayList<>(
    List.of(127, 128, 127, 128));

int s = 0;
for (int i = 0; i < a.size(); i++)
    if (a.get(i) == b.get(i)) s++;

a.add(a.remove(0));
a.remove(Integer.valueOf(128));
b.subList(0, 2).clear();
out.println(s + " " + a + " " + b);
```

Question 16

What is output by the code to the right?

A) `1213` **B)** `13` **C)** `123`

D) There is no output due to a compile error.

E) There is no output due to a runtime error.

```
try {
    String str = "12\n34";
    Scanner s = new Scanner(str);
    out.print(s.nextLine());
} catch(IOException f) {
    out.print(1);
} finally {
    out.print(3);
}
```

Question 17

What is output by the line marked `//q17` in the code to the right?

A) `[A, B, B, E, E, F, H, J, K, L, S]`

B) `[A, B, E, J, F, K, S, H, L, E, B]`

C) `[A, B, E, F, H, J, K, L, S]`

D) `[A, B, E, J, F, K, S, H, L]`

E) There is no output due to a runtime error.

Question 18

Assuming any error in the preceding code has been corrected, which of the following values could **not** possibly be output by the line marked `//q18` in the code to the right?

A) `0` **B)** `1` **C)** `98` **D)** `99`

E) More than one of the above.

Question 19

Let N represent an arbitrary length of `String s`. In terms of N, what is the Big-$\mathcal{O}$ runtime of the code to the right (assume average-case)?

A) $\mathcal{O}(N \log N)$ **B)** $\mathcal{O}(N)$

C) $\mathcal{O}(N^2)$ **D)** $\mathcal{O}(\log N)$

E) $\mathcal{O}(N^2 \log N)$

```
TreeMap<String,Integer> tm;
tm = new TreeMap<>();

String[] s ="ABEJFKSHLEB".split("");
for(String r:s) {
    int num =(int)(Math.random() * 99);
    tm.put(r, num);
}

out.println(tm.keySet()); //q17

for(String h:tm.keySet())
    out.print(tm.get(h)); //q18
```

Question 20

Which of the lines in the code to the right first causes an error (compile- or run-time)?

A) `//1` **B)** `//2` **C)** `//3` **D)** `//4`

E) None of the above. All lines are error-free.

```
int[][] m = new int[5][]; //1
m[3] = new int['A'];      //2
out.print(m.hashCode());  //3
m[2][0] = 17;             //4
```

Question 21

What is output by the code to the right?

A) `2570` **B)** `1400` **C)** `1825`

D) There is no output due to a compile error.

E) There is no output due to a runtime error.

```
String s = "30201";
int i = Integer.parseInt(s, 5);

s = Integer.toString(i, 9);
i = Integer.parseInt(s, 7);
s = Integer.toString(i, 9);
i = Integer.parseInt(s, 7);

out.println(i);
```

Question 22

What can replace **<1*>** in the code to the right so that it compiles and runs without error?

A) `extends Fighter`

B) `implements Fighter`

C) `extends Wizard, Fighter`

D) `extends Warrior`

E) More than one of the above.

Question 23

What can replace **<2*>** in the code to the right so that the `Wizard()` constructor initializes the `health` instance variable to `200`, and the `magic` instance variable to the parameter `m`?

A)
```
super();
magic = m;
```

B)
```
magic = m;
```

C)
```
super(200);
magic = m;
```

D)
```
super(200);
```

E) A or B.

F) A, B, or C.

G) Any of the above.

H) None of the above.

Question 24

Assuming **<1*>** and **<2*>** are filled in correctly, what is output by the line marked `//q24` in the code to the right?

A) `200`

B) `180`

C) `170`

D) `160`

E) There is no output due to a runtime error.

Question 25

Assuming **<1*>** and **<2*>** are filled in correctly, what is output by the line marked `//q25` in the code to the right?

A) `260 23`

B) `200 23`

C) `200 25`

D) `180 23`

E) There is no output due to a runtime error.

Question 26

Assuming **<1*>** and **<2*>** are filled in correctly, what is output by the line marked `//q26` in the code to the right?

A) `320`

B) `170`

C) `250`

D) `350`

E) There is no output due to a runtime error.

```
class Fighter{
    private int health;
    public Fighter() {
        health = 200;
    }
    public int getHealth() {
        return health;
    }
    public void setHealth(int h) {
        health = h;
    }
    public int attack(int eh) {
        return eh - 10;
    }
}

class Wizard <1*> {
    int magic;
    public Wizard(int m) {
       <2*>;
    }
    public int attack(int eh) {
        magic--;
        return eh - 20;
    }
}

class Warrior <1*> {
    public int health;
    public Warrior() {
        health = 300;
    }
    public void heal(){
        setHealth(health + 50);
    }
}

////////////// client code ///////////
Fighter f = new Fighter();
Wizard w1 = new Wizard(25);
Warrior w2 = new Warrior();

int num = w1.attack(f.getHealth());
f.setHealth(num);
num = w2.attack(f.getHealth());
f.setHealth(num);
num = w1.attack(w2.getHealth());
w2.setHealth(num);
num = f.attack(w2.getHealth());
w2.setHealth(num);
w2.heal();

out.println(f.getHealth());    //q24
String o = w1.getHealth()+" ";
out.println(o + w1.magic);     //q25
out.println(w2.getHealth());   //q26
```

Question 27

What is output by the line marked `//q28` in the code to the right?

A) `false false` **B)** `false true`

C) `true false` **D)** `true true`

E) There is no output due to a compile error.

```
String a = "3LF BO0 0n3 TW0 3E3";

String reg = "((\\d[A-Z][A-Z])|";
reg += "(\\w\\D\\S) ?){4,6}";
out.print(a.matches(reg));

out.print(" ");
reg = ".+((?<=([e-x]))[2-8]).*";

reg += "\\s.*((?<!(\\s))\\d).*";
out.println(a.matches(reg));
```

Question 28

What is output by the line marked `//q28` in the code to the right?

A) `14` **B)** `13`

C) `22` **D)** `18`

E) There is no output due to a runtime error.

Question 29

What is output by the line marked `//q29` in the code to the right?

A) `141` **B)** `128`

C) `376` **D)** `484`

E) There is no output due to a runtime error.

```
public int recur(int n) {
    if(n <= 0) return 1;
    if(n % 3 == 1)
        return 2 * recur(n - 4);
    if(n < 10)
        return n + recur(n / 2);
    if(n % 7 < 2)
        return 3 + recur(n - 4);
    return recur(n - 6) + recur(n - 8);
}

////////////// client code ////////////
out.println(recur(13));   //q28
out.println(recur(43));   //q29
```

Question 30

From which two sorting algorithms is the TIMSORT algorithm derived (commonly used in the `Collections.sort()` and `Arrays.sort()` methods)?

A) SELECTIONSORT and INSERTIONSORT **B)** INSERTIONSORT and QUICKSORT **C)** SELECTIONSORT and QUICKSORT

D) INSERTIONSORT and MERGESORT **E)** SELECTIONSORT and MERGESORT **F)** QUICKSORT and MERGESORT

Question 31

Which of the following Java operators are right-to-left associative according to standard Java operator precedence rules?

A) 1, 2, and 5 only. **B)** 2, 6, 7, and 8 only.

C) 2, 3, 6, 7, and 8 only. **D)** 4, 6, 7, and 8 only.

E) None of the above.

```
1. new          2. ++ (post-increment)
3. !=           4. ++ (pre-increment)
5. ==           6. =
7. <<=          8. -> (lambda expression)
```

Question 32

What is output by the client code to the right?

A) `AB` **B)** `ABAB`

C) `ABCA` **D)** `BCB C`

E) There is no output due to an error.

```
UnaryOperator<String> shift = s ->
        s.substring(1) + s.charAt(0);
Predicate<String> p = s ->
        s.compareTo("BCA") >= 0;

String r = Stream.iterate("ABC", shift)
        .limit(5)
        .filter(p.negate())
        .map(s -> s.substring(0, 2))
        .collect(Collectors.joining());


out.println(r);
```

Question 33

What is output by the client code to the right?

A)
```
Beginner
true
MEDIUM
null
```

B)
```
EASY
true
2
null
```

C)
```
Beginner
false
MEDIUM
Exception
```

D)
```
Beginner
true
Intermediate
null
```

E) There is no output due to an error.

Question 34

Which of the following statements about the `Difficulty` enum is true?

A) Enums cannot have constructors.

B) The `values()` method must be manually implemented.

C) Each enum constant (`EASY`, `MEDIUM`, `HARD`) is an instance of `Difficulty`.

D) Enum fields must be declared as `static`.

E) More than one of the above is true.

Question 35

What would the following expression evaluate to?

```
Difficulty.MEDIUM.isHarderThan(Difficulty.HARD)
```

A) `true`

B) `false`

C) There is no output due to a compile error.

D) There is no output due to a runtime error.

```
enum Difficulty {
    EASY(1, "Beginner"),
    MEDIUM(2, "Intermediate"),
    HARD(3, "Advanced");

    private int level;
    private String description;

    Difficulty(int level, String description) {
        this.level = level;
        this.description = description;
    }

    public int getLevel() {
        return level;
    }

    public String getDescription() {
        return description;
    }

    public boolean isHarderThan(Difficulty other) {
        return this.level > other.level;
    }

    public static Difficulty fromLevel(int level) {
        for (Difficulty d : values()) {
            if (d.level == level) {
                return d;
            }
        }
        return null;
    }
}

////////////////// Client Code /////////////////
Difficulty d1 = Difficulty.EASY;
Difficulty d2 = Difficulty.HARD;

out.println(d1.getDescription());
out.println(d2.isHarderThan(d1));
out.println(Difficulty.fromLevel(2));
out.println(Difficulty.fromLevel(4));
```

Question 36

What is output by the code to the right?

A) `+A+BX-A-B`

B) `+A+BX-B-A`

C) `+A-B+BX-A`

D) `+B+AX-A-B`

E) The output cannot be determined until runtime.

```
static class R implements AutoCloseable {
    private String name;

    R(String n) {
        name = n;
        out.print("+" + name);
    }

    public void close() {
        out.print("-" + name);
    }
}

////////////// Client Code //////////////
try (R a = new R("A"); R b = new R("B")) {
    out.print("X");
}
```

Question 37

What does the method `mysteryFunc1()` to the right compute?

A) The sum of all elements in `nums`.

B) The maximum element in `nums`.

C) The maximum subarray sum of `nums`.

D) The longest increasing subsequence in `nums`.

E) A needlessly complicated way of returning the first element in `nums`.

Question 38

Suppose the method `mysteryFunc1()` was called with an array of n integers. What would be the time complexity of said method call?

A) $\mathcal{O}(n)$

B) $\mathcal{O}(n \log n)$

C) $\mathcal{O}(n^2)$

D) $\mathcal{O}(\log n)$

E) $\mathcal{O}(n \log^2 n)$

```
private static class Tuple {
    int a, b, c, d;
}

private Tuple helper1(int[] arr, int l, int h) {
    if (l == h) {
        int val = arr[l];
        Tuple res = new Tuple();
        res.a = res.b = res.c = res.d = val;
        return res;
    }

    int m = (l + h) / 2;
    Tuple L = helper1(arr, l, m);
    Tuple R = helper1(arr, m + 1, h);
    return helper2(L, R);
}

private Tuple helper2(Tuple L, Tuple R) {
    Tuple res = new Tuple();
    res.a = L.a + R.a;
    res.b = Math.max(L.b, L.a + R.b);
    res.c = Math.max(R.c, L.c + R.a);
    res.d = Math.max(Math.max(L.d, R.d),
                     L.c + R.b);
    return res;
}

public int mysteryFunc1(int[] nums) {
    if (nums.length == 0) return 0;
    return helper1(nums, 0, nums.length - 1).d;
}
```

Question 39

Consider the weighted graph G to the right. What is the total weight of a minimum spanning tree of graph G? Write your answer in the blank provided on your answer sheet.

Question 40

Consider the weighted graph G to the right. How many unique shortest path traversals exist starting from vertex v_0 to vertex v_6? Write your answer in the blank provided on your answer sheet.

